

PROBLEM STATEMENT

Predictive Analytics System for Early Detection of Land Acquisition Delays

Industry Context

Land acquisition is an important part of infrastructure and development projects such as highways, railways, industrial corridors, airports, dams, and urban development. Large projects often require acquiring land from multiple owners and completing several administrative, legal, financial, and rehabilitation procedures.

Land acquisition processes can experience delays due to property ownership disputes, incomplete documentation, compensation issues, legal cases, objections from landowners, administrative bottlenecks, and delays in government approvals. These delays can affect the overall project schedule and increase construction costs.

Traditional land acquisition monitoring systems generally focus on tracking the current status of cases rather than **predicting future delays**. As a result, authorities and project managers may identify critical problems only after significant time has already been lost. There is a need for an intelligent predictive system that can analyze historical and current acquisition data and identify cases that are likely to experience delays.

Problem Statement

Design and develop a Predictive Analytics System for Early Detection of Land Acquisition Delays that analyzes land acquisition records, project information, legal status, documentation status, compensation details, approval stages, and historical processing data.

The system should collect and process relevant land acquisition data from different stages of the acquisition process. Using data analytics and machine learning techniques, the system should identify patterns and factors associated with delayed acquisitions and predict the likelihood of delay for individual land parcels or acquisition cases.

The platform should provide early warnings and risk indicators to government authorities, project managers, and other authorized stakeholders when a land acquisition case shows a high possibility of delay. This will allow stakeholders to take preventive actions such as resolving

documentation issues, addressing compensation concerns, prioritizing approvals, or identifying cases requiring additional attention.

The system should also provide dashboards, visual analytics, historical reports, and case-level insights to improve transparency and decision-making throughout the land acquisition process.

Objectives

The system should:

- To Collect and organize land acquisition data from multiple stages.
- To Analyze historical land acquisition records to identify delay patterns.
- To Identify major factors contributing to acquisition delays.
- To Predict the possibility of future delays at an early stage.
- To Generate alerts for high-risk land acquisition cases.
- To Provide risk indicators for individual land parcels or cases.
- To Help authorities prioritize cases requiring immediate attention.
- To Provide dashboards and visual reports for project monitoring.
- To Maintain historical records and audit information.
- To Improve planning and decision-making for infrastructure projects.

Expected Outcome

1. Develop an end-to-end Predictive Analytics System for collecting, processing, analyzing, and predicting land acquisition delays.
2. Provide a GUI/dashboard for monitoring land acquisition cases, viewing delay risks, and visualizing project progress.
3. Maintain information such as land parcel details, ownership status, documentation status, compensation status, legal issues, approval stages, acquisition progress, historical delays, and project details.
4. Apply data analytics and machine learning techniques to identify patterns and factors associated with land acquisition delays.
5. Predict the likelihood of delay for individual land acquisition cases and classify them based on their risk level.
6. Generate early-warning alerts and notifications for cases that have a high possibility of experiencing delays.
7. Provide historical analytics, reports, and visualizations to support project managers and authorized authorities in making data-driven decisions.

8. Apply key software engineering principles such as security, scalability, maintainability, reliability, usability, and data privacy, while demonstrating proper requirements analysis, system design, SDLC selection, risk evaluation, limitations, and future enhancements.